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- P. fraction

CHAPTER- 4- cont.

Partial fraction expansion method

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4.3 The inverse Z-transform – **cont.**

4.3.2 Partial fraction expansion method

Sec. 4.3.2

4.3.2 Partial fraction expansion method

- In this method, the z-transform is first expanded into a sum of simple partial fractions.

- The **IZT** of each partial fraction is then obtained from tables, such as **Table 4.1**, and then summed to give the overall **IZT.**

Sec. 4.3 Cont. - 6

Table 4.1 z-transforms .

Entry number	*Discrete-time sequence x(n), n ≥ 0*	*z-transform X(z)*	*Region of convergence of X(z)*
1	$k\delta(n)$	k	Everywhere
2	K	$\frac{kz}{z-1}$	|z| > 1
3	kn	$\frac{kz}{(z-1)^2}$	|z| > 1
4	kn^2	$\frac{kz(z+1)}{(z-1)^3}$	|z| > 1
5	$Ke^{-\alpha n}$	$\frac{kz}{z-e^{-\alpha}}$	|z| > $e^{-\alpha}$
6	$Kne^{-\alpha n}$	$\frac{kze^{-\alpha}}{(z-e^{-\alpha})^2}$	|z| > $e^{-\alpha}$
7	$1-e^{-\alpha n}$	$\frac{z(1-e^{-\alpha})}{z^2-z(1+e^{-\alpha})+e^{-\alpha}}$	|z| > $e^{-\alpha}$
8	$\cos(\alpha n)$	$\frac{z(z-\cos\alpha)}{z^2-2z\cos\alpha+1}$	|z| > 1

k and are constants; c is a complex number.

Sec. 4.3 Cont. - 7

Table 4.1 z-transforms - **Cont.**

9	$\sin(\alpha n)$	$\frac{z\sin\alpha}{z^2-2z\cos\alpha+1}$	$\lvert z\rvert > 1$
10	$E^{-\alpha n}\sin(\alpha n)$	$\frac{ze^{-\alpha}\sin\alpha}{z^2-2e^{-\alpha}z\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
11	$e^{-\alpha n}\cos(\alpha n)$	$\frac{ze^{-\alpha}(ze^{\alpha}-\cos\alpha)}{z^2-2ze^{-\alpha}\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
12	$\cosh(\alpha n)$	$\frac{z^2-z\cosh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \cosh\alpha$
13	$\sinh(\alpha n)$	$\frac{z\sinh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \sinh\alpha$
14	$k\alpha^n$	$\frac{kz}{z-\alpha}$	$\lvert z\rvert > \alpha$
15	$kn\alpha^n$	$\frac{k\alpha z}{(z-\alpha)^2}$	$\lvert z\rvert > \alpha$
16	$2\lvert c\rvert\,\lvert p\rvert^n\cos(n\angle p+\angle c)$	$\frac{cz}{z-p}+\frac{c^*z}{z-p^*}$	

k and are constants; c is a complex number.

Sec. 4.3.2 Cont. - 1

- In many practical cases, the z-transform is given as a ratio of polynomials in **z** or **Z^{-1}** and has the following familiar form, **Eq. 4.9**.

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}} \qquad (4.9)$$

Sec. 4.3.2 Cont. - 2

- If the poles of X(z) are of first order and **N=M,** then X(z) can be expanded as:

$$X(z) = B_0 + \frac{C_1}{1-P_1 z^{-1}} + \frac{C_2}{1-P_2 z^{-1}} + \ldots\ldots + \frac{C_M}{1-P_M z^{-1}}$$

$$= B_0 + \frac{C_1 z}{z-P_1} + \frac{C_2 z}{z-P_2} + \ldots + \frac{C_M z}{z-P_M}$$

$$= B_0 + \sum_{k=1}^{M} \frac{C_k z}{z-P_k} \qquad (4.10)$$

Sec. 4.3.2 Cont. - 3

Where; P_k are the poles of X(z) (assumed distinct), and

C_k are the partial fraction coefficients and

$$B_0 = a_N / b_M \qquad (4.11)$$

The C_k are also known as the residues of X(z); **see section 4.3.3.**

Sec. 4.3.2 Cont. - 4

- If **N < M**, in **Eq. 4.9**, then **B_0** will be zero. Then, X(z) as **Eq. 4.9- 1**

$$X(z) = \sum_{k=1}^{M} \frac{C_k z}{z - P_k} \qquad (4.9\text{-}1)$$

- If **N > M** then X(z) must be reduced first, to make **$N \leq M$**, by long division with the numerator and denominator polynomials written in descending powers of **z^{-1}** .

- The remainder can then be expressed as in **Eq. 4.10.**

$$X(z) = B_0 + \sum_{k=1}^{M} \frac{C_k z}{z - P_k} \qquad (4.10)$$

Sec. 4.3.2 Cont. - 5

The coefficient, C_k, associated with the pole P_k may be obtained by multiplying both sides of **Eq. 4.10** by $(z - P_k)/z$ and then letting $z = P_k$, as shown in **Eq. 4.12:**

$$C_k = \left. \frac{X(z)}{z}(z - p_k) \right|_{z=p_1} \qquad (4.12)$$

Sec. 4.3.2 Cont. - 6

Special case:

- If X(z) contains one or more multiple-order poles (**i. e.** poles that are coincident) then extra terms are required in **Eq. 4.10** to take this into account.

- For example, if X(z) contains an **m^{th}**-order pole at **$z = P_k$** the partial fraction expansion must include terms of the form:

$$\sum_{i=1}^{m} \frac{D_i}{(z - p_k)^i}$$

(4.13a)

Sec. 4.3.2 Cont. - 7

$$\sum_{i=1}^{m} \frac{D_i}{(z - p_k)^i} \qquad \textbf{(4.13a)}$$

The coefficients, D_i, may be obtained from the relationship:

$$D_i = \frac{1}{(m-i)!} \frac{d^{m-i}}{dz^{m-i}} \left[\frac{(z - p_k)^m X(z)}{z} \right]_{z=p_k} \qquad \textbf{(4.13b)}$$

Evaluation of **IZT** by the partial fraction expansion method **is best illustrated by examples**.

Sec. 4.3.2 Cont. - 8

Example 4.4 **(X(z) contains simple, first-order poles)**

Find the inverse z-transform of the following:

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}}$$

Solution

For simplicity, we first express the z-transform in positive powers of z by multiplying the numerator and denominator by z^2 (**the highest power of z**):

$$X(z) = \frac{z}{z^2 - 0.25z - 0.375} = \frac{z}{(z - 0.75)(z + 0.5)}$$

Sec. 4.3.2 Cont. - 9

- X(z) Contains first-order poles at z = 0.75 and at z = - 0.5 (**i. e.,** only one pole occurs at each pole position).

- Since **N<M,** the partial fraction expansion has the form:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)} = \frac{C_1 z}{z-0.75} + \frac{C_2 z}{z+0.5} \qquad \textbf{(4.14)}$$

Sec. 4.3.2 Cont. - 10

- To make it easier to find the values of the C_k we divide both sides by **z, Eq. 4.15**:

$$\frac{X(z)}{z} = \frac{z}{z(z-0.75)(z+0.5)} = \frac{C_1}{z-0.75} + \frac{C_2}{z+0.5} \qquad (4.15)$$

- **To obtain C_1,** we simply multiply both sides of **Eq. 4.15** by **(z-0.75)** and let **z = 0.75** :

$$\frac{(z-0.75)X(z)}{z} = \frac{(z-0.75)}{(z-0.75)(z+0.5)} = C_1 + \frac{C_2(z-0.75)}{z+0.5}$$

$$C_1 = \left.\frac{1}{z+0.5}\right|_{z=0.75} = \frac{1}{0.75+0.5} = \frac{4}{5}$$

Sec. 4.3.2 Cont. - 11

- **Similarly**, C_2, is obtained as:

$$\frac{(z-0.5)X(z)}{z} = \frac{(z-0.5)}{z-0.75}C_1 + C_2\frac{(z-0.5)}{z+0.5}$$

$$C_2 = \left.\frac{(z+0.5)X(z)}{z}\right|_{z=-0.5}$$

$$= \left.\frac{(z+0.5)}{(z-0.75)(z+0.5)}\right|_{z=-0.5} = \frac{1}{-0.5-0.75} = -\frac{4}{5}$$

Sec. 4.3.2 Cont. - 12

- Using the values of C_1 and C_2 in **Eq. 4.14** we have:

$$X(z) = \frac{(4/5)z}{z-0.75} - \frac{(4/5)z}{z+0.5} \qquad \textbf{(4.16)}$$

- From the z-transform table, **table 4.1, entry 14,** the IZT of each term on the right-hand side of **Eq. 4.16** is given as:

$$z^{-1}\left[\frac{(4/5)z}{z-0.75}\right] = \frac{4(0.75)^n}{5}$$

$$z^{-1}\left[\frac{-(4/5)z}{z+0.5}\right] = \frac{-4(-0.5)^n}{5}$$

Sec. 4.3 Cont. - 6

Table 4.1 z-transforms .

Entry number	*Discrete-time sequence x(n), n ≥ 0*	*z-transform X(z)*	*Region of convergence of X(z)*
1	$k\delta(n)$	k	Everywhere
2	K	$\frac{kz}{z-1}$	|z| > 1
3	kn	$\frac{kz}{(z-1)^2}$	|z| > 1
4	kn^2	$\frac{kz(z+1)}{(z-1)^3}$	|z| > 1
5	$Ke^{-\alpha n}$	$\frac{kz}{z-e^{-\alpha}}$	|z| > $e^{-\alpha}$
6	$Kne^{-\alpha n}$	$\frac{kze^{-\alpha}}{(z-e^{-\alpha})^2}$	|z| > $e^{-\alpha}$
7	$1\text{-}e^{-\alpha n}$	$\frac{z(1-e^{-\alpha})}{z^2-z(1+e^{-\alpha})+e^{-\alpha}}$	|z| > $e^{-\alpha}$
8	$\cos(\alpha n)$	$\frac{z(z-\cos\alpha)}{z^2-2z\cos\alpha+1}$	|z| > 1

k and are constants; c is a complex number.

Sec. 4.3 Cont. - 7

Table 4.1 z-transforms - **Cont.**

9	$\sin(\alpha n)$	$\frac{z\sin\alpha}{z^2-2z\cos\alpha+1}$	$\lvert z\rvert > 1$
10	$E^{-\alpha n}\sin(\alpha n)$	$\frac{ze^{-\alpha}\sin\alpha}{z^2-2e^{-\alpha}z\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
11	$e^{-\alpha n}\cos(\alpha n)$	$\frac{ze^{-\alpha}(ze^{\alpha}-\cos\alpha)}{z^2-2ze^{-\alpha}\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
12	$\cosh(\alpha n)$	$\frac{z^2-z\cosh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \cosh\alpha$
13	$\sinh(\alpha n)$	$\frac{z\sinh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \sinh\alpha$
14	$k\alpha^n$	$\frac{kz}{z-\alpha}$	$\lvert z\rvert > \alpha$
15	$kn\alpha^n$	$\frac{k\alpha z}{(z-\alpha)^2}$	$\lvert z\rvert > \alpha$
16	$2\lvert c\rvert\,\lvert p\rvert^n\cos(n\angle p+\angle c)$	$\frac{cz}{z-p}+\frac{c^*z}{z-p^*}$	

k and are constants; c is a complex number.

Sec. 4.3.2 Cont. - 13

- The desired IZT, x(n), is the sum of the two **IZTs**:

$$X(n) = \frac{4}{5}[(0.75)^n - (-0.5)^n],\ n > 0$$

Example 4.6 (X(z) contains a second-order pole)

Find the discrete-time sequence, **x(n)**, with the following z-transform :

$$X(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

Solution

- X(z) has a first-order pole at z = 0.5 and a second-order pole at z = 1.
- In this case, the partial fraction expansion has the form:

$$X(z) = \frac{CZ}{z-0.5} + \frac{D_1 Z}{z-1} + \frac{D_2 Z}{(z-1)^2} \qquad (4.21)$$

Sec. 4.3.2 Cont. - 21

$$X(z) = \frac{CZ}{z-0.5} + \frac{D_1 Z}{z-1} + \frac{D_2 Z}{(z-1)^2} \qquad (4.21)$$

- To obtain **C**, we proceed as before and multiply both sides of **Eq. 4.21** by **(z–0.5)**, set **z = 0.5** and evaluate the expression:

$$C = \left.\frac{(z-0.5)z^2}{z(z-0.5)(z-1)^2}\right|_{z=0.5} = 0.5/(0.5\text{-}1)^2 = 2$$

Sec. 4.3.2 Cont. - 22

- To obtain D_1 we use **Eq. 4.13b**, with **i = 1 and m = 2. Thus:**

$$D_1 = \frac{d}{dz}\left[\frac{(z-1)^2 X(z)}{z}\right]_{z=1} = \frac{d}{dz}\left[\frac{(z-1)^2 z^2}{z(z-0.5)(z-1)^2}\right]_{z=1}$$

$$= \frac{d}{dz}\left[\frac{z}{(z-0.5)}\right]_{z=1} = \left.\frac{z-0.5-z}{(z-0.5)^2}\right|_{z=1}$$

$$= \left.\left(\frac{-0.5}{z-0.5}\right)\right|_{z=1} = \frac{-0.5}{(0.5)^2} = \frac{-0.5}{0.5\times 0.5} = -2$$

Sec. 4.3.2 Cont. - 23

- Similarly, D_2 is obtained from **Eq. 4.13b** by **letting i = m = 2**

$$D_2 = \frac{1}{(m-i)} \frac{d^{m-i}}{dz^{m-i}} \left[(z - P_k)^m X(z) \right]_{z=P_k}$$

$$= \left(\frac{(z-1)^2 X(z)}{z} \right)\Bigg|_{z=1} = \left(\frac{(z-1)^2 z^2}{z(z-0.5)(z-1)^2} \right)\Bigg|_{z=1}$$

$$= \left[\frac{z}{z-5} \right]_{z=1} = \frac{1}{1-0.5} = 2$$

- Combining the results, X(z) becomes:

$$X(z) = \frac{2z}{z-0.5} - \frac{2z}{z-1} + \frac{2z}{(z-1)^2}$$

Sec. 4.3.2 Cont. - 24

$$X(z) = \frac{2z}{z-0.5} - \frac{2z}{z-1} + \frac{2z}{(z-1)^2}$$

- The **IZT** of each term on the right-hand side is obtained from **Table 4.1** and then summed to give **x(n)** :

$$x(n) = 2(0.5)^n - 2 + 2n = 2[(n-1)+(0.5)^n],\ n \geq 0 \qquad (4.22)$$

The reader can verify that the result is correct by comparing the first few values of x(n) with values calculated with the power series method.

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- Residue

CHAPTER- 4- cont.

Residue method

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4.3 The inverse Z-transform – **cont.**

4.3.3 Residue method

4.3.4 Comparison of inverse z-tran. methods

Sec. 4.3.3

4.3.3 Residue method

- In this method the IZT is obtained by evaluating the contour integral

$$x(n) = \frac{1}{2\pi j} \oint_C z^{n-1} X(z)\, dz \qquad (4.23)$$

- where C is the path of integration enclosing all the poles of X(z).